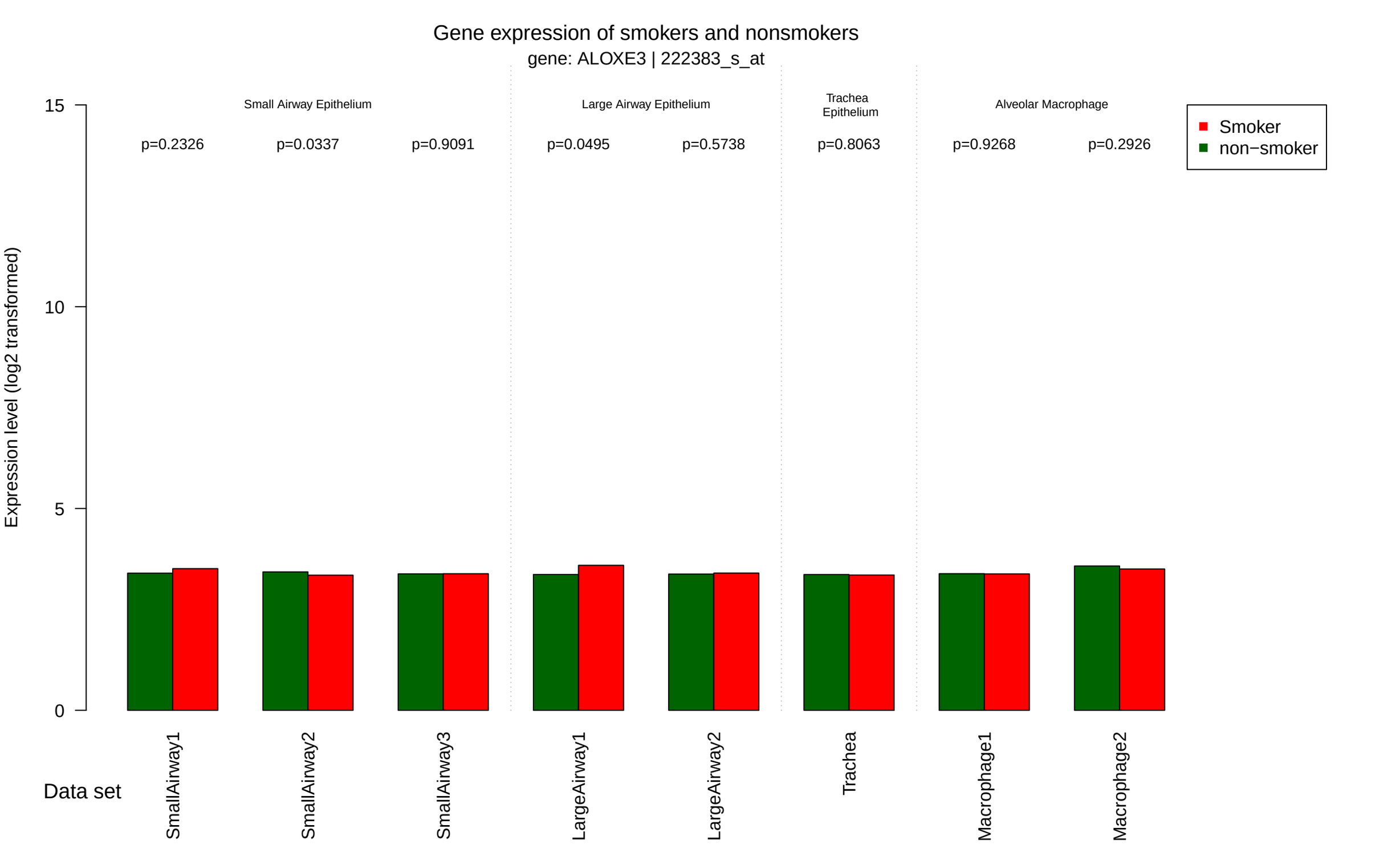
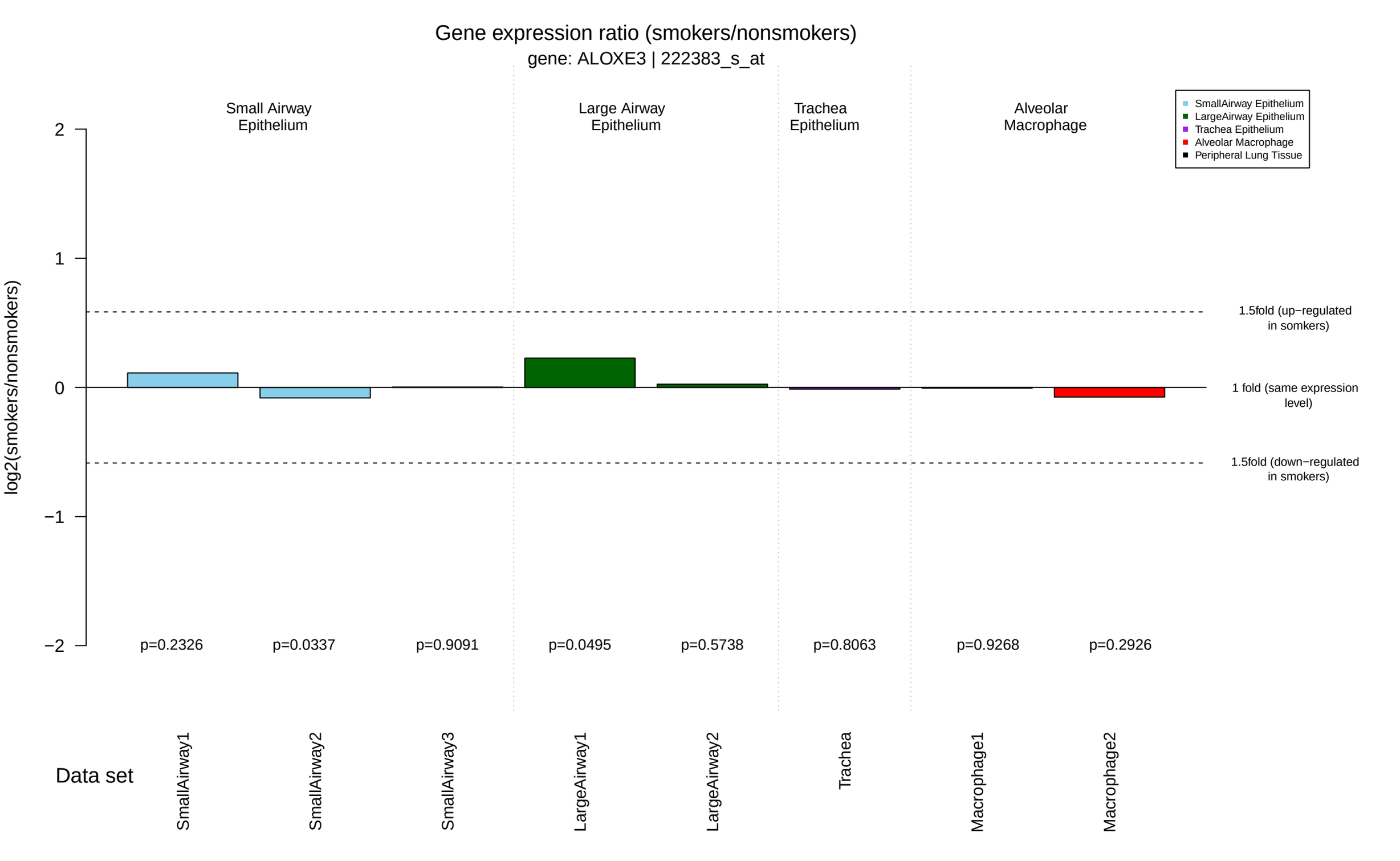
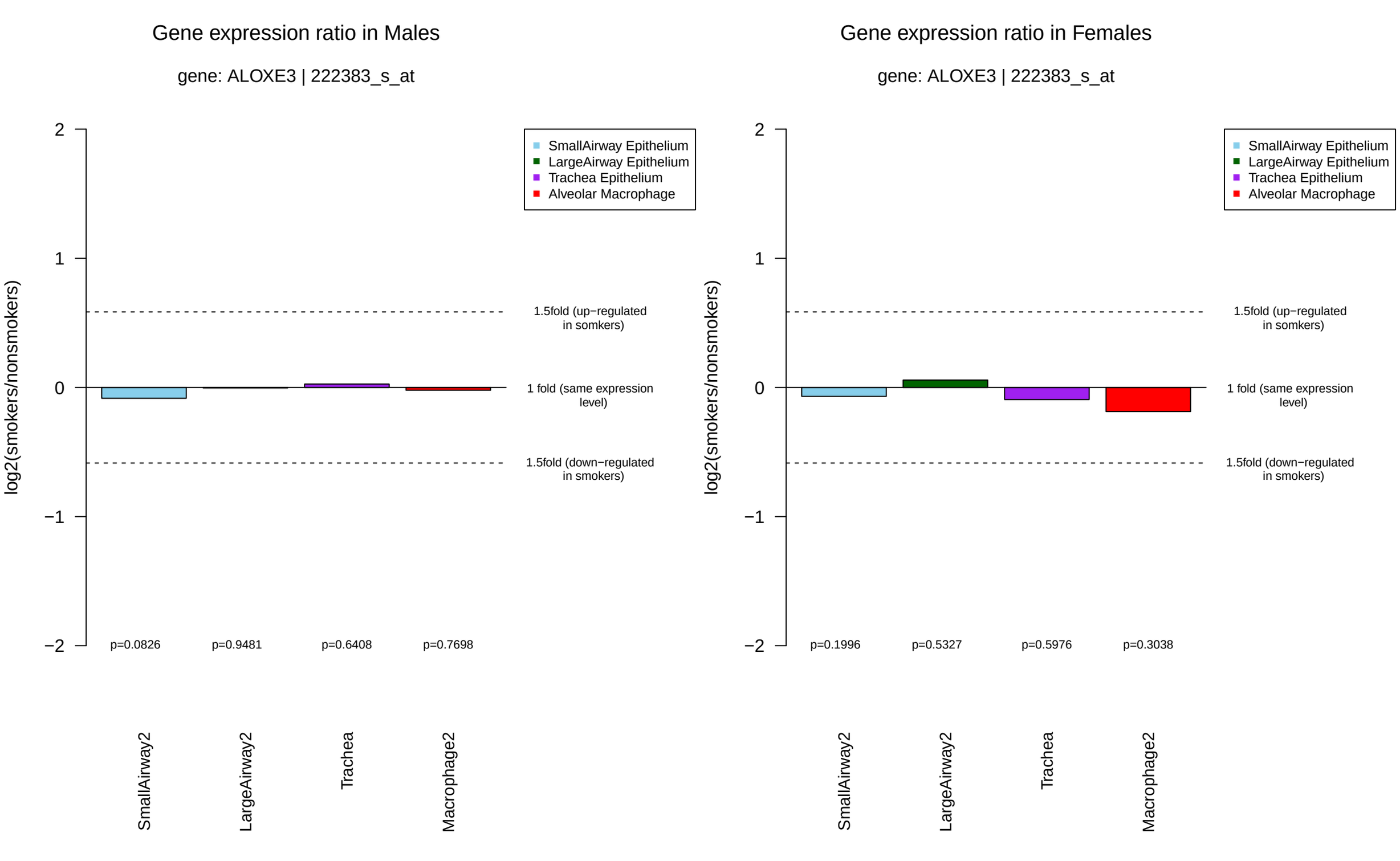
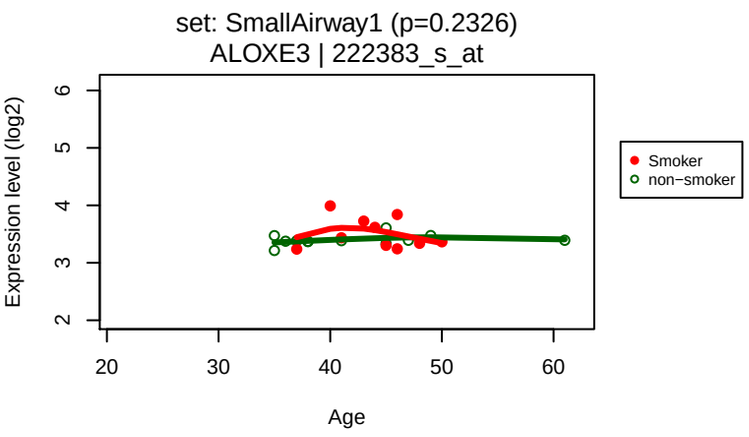
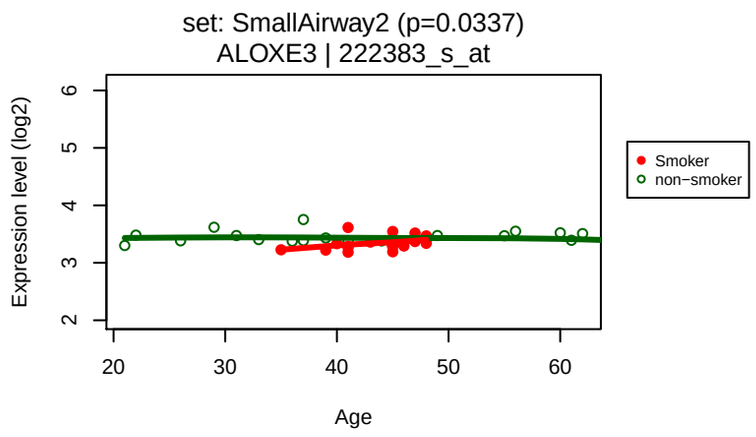
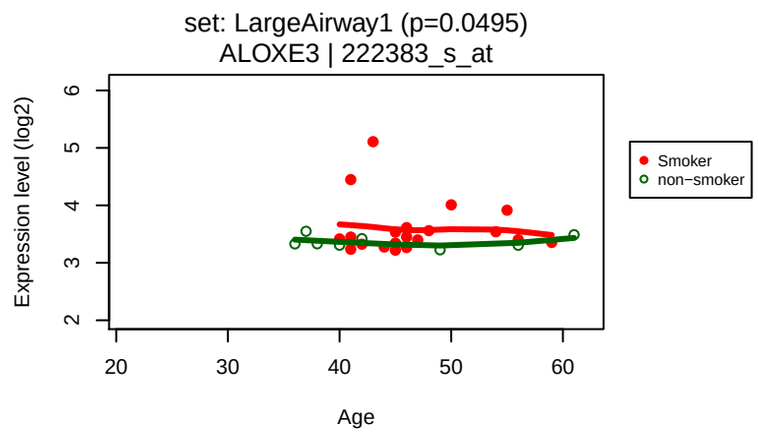

Gene expression of smokers and nonsmokers
gene: ALOXE3 | 222383_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2326
p=0.0337
p=0.9091
p=0.0495
p=0.5738
p=0.8063
p=0.9268
p=0.2926
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

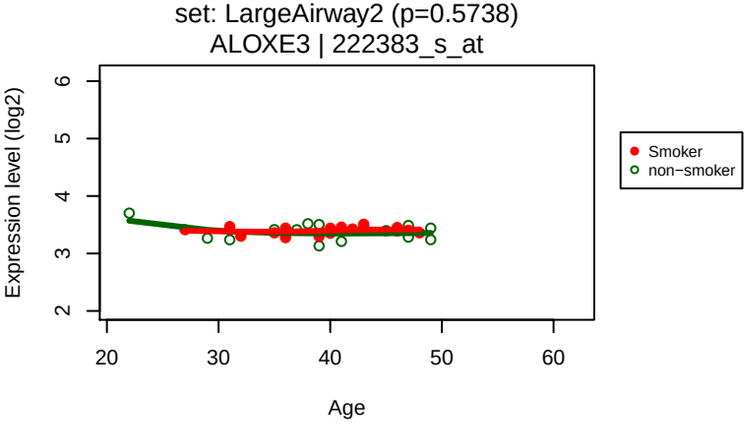

Gene expression ratio (smokers/nonsmokers)
gene: ALOXE3 | 222383_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2326
p=0.0337
p=0.9091
p=0.0495
p=0.5738
p=0.8063
p=0.9268
p=0.2926
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

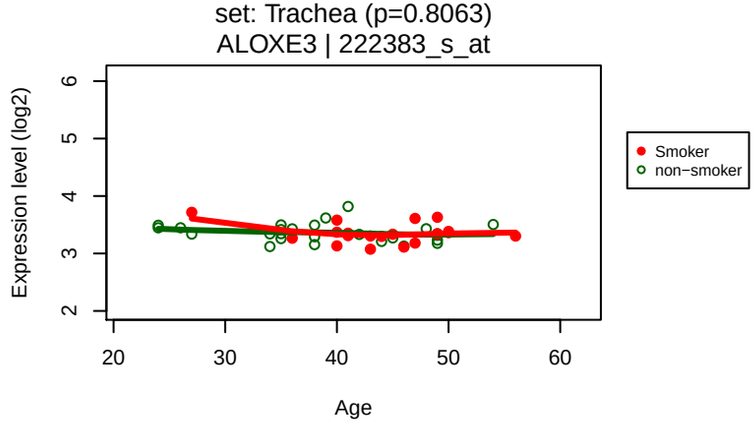

Gene expression ratio in Males
gene: ALOXE3 | 222383_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0826
p=0.9481
p=0.6408
p=0.7698
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOXE3 | 222383_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1996
p=0.5327
p=0.5976
p=0.3038
SmallAirway2
LargeAirway2
Trachea
Macrophage2

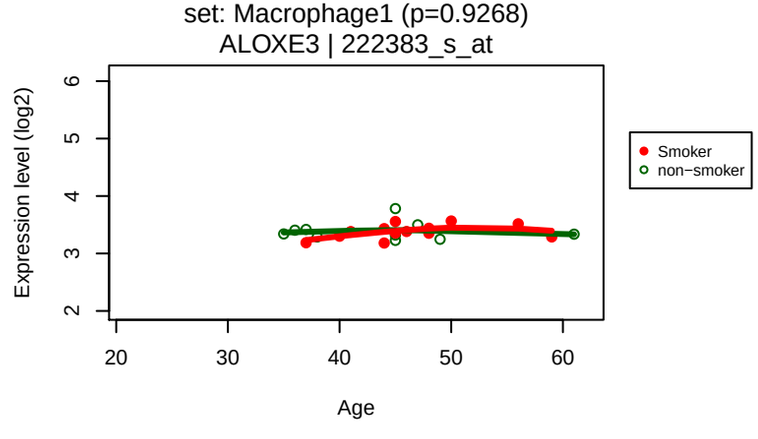

set: SmallAirway1 (p=0.2326)
ALOXE3 | 222383_s_at
Expression level (log2)
Age
Smoker
non−smoker

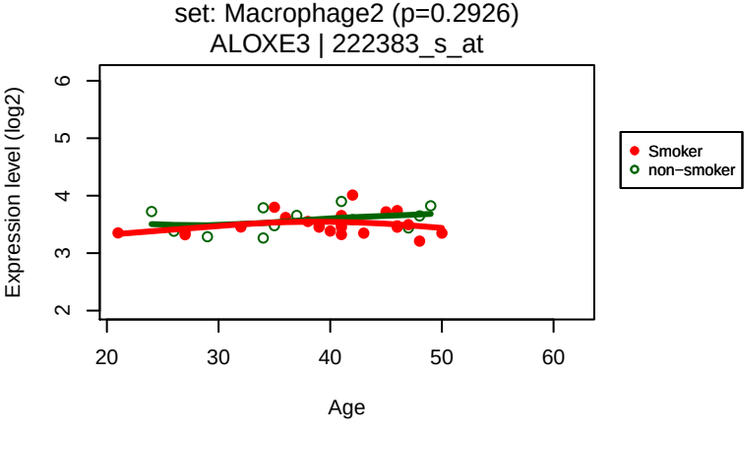

set: SmallAirway2 (p=0.0337)
ALOXE3 | 222383_s_at
Expression level (log2)
Age
Smoker
non−smoker

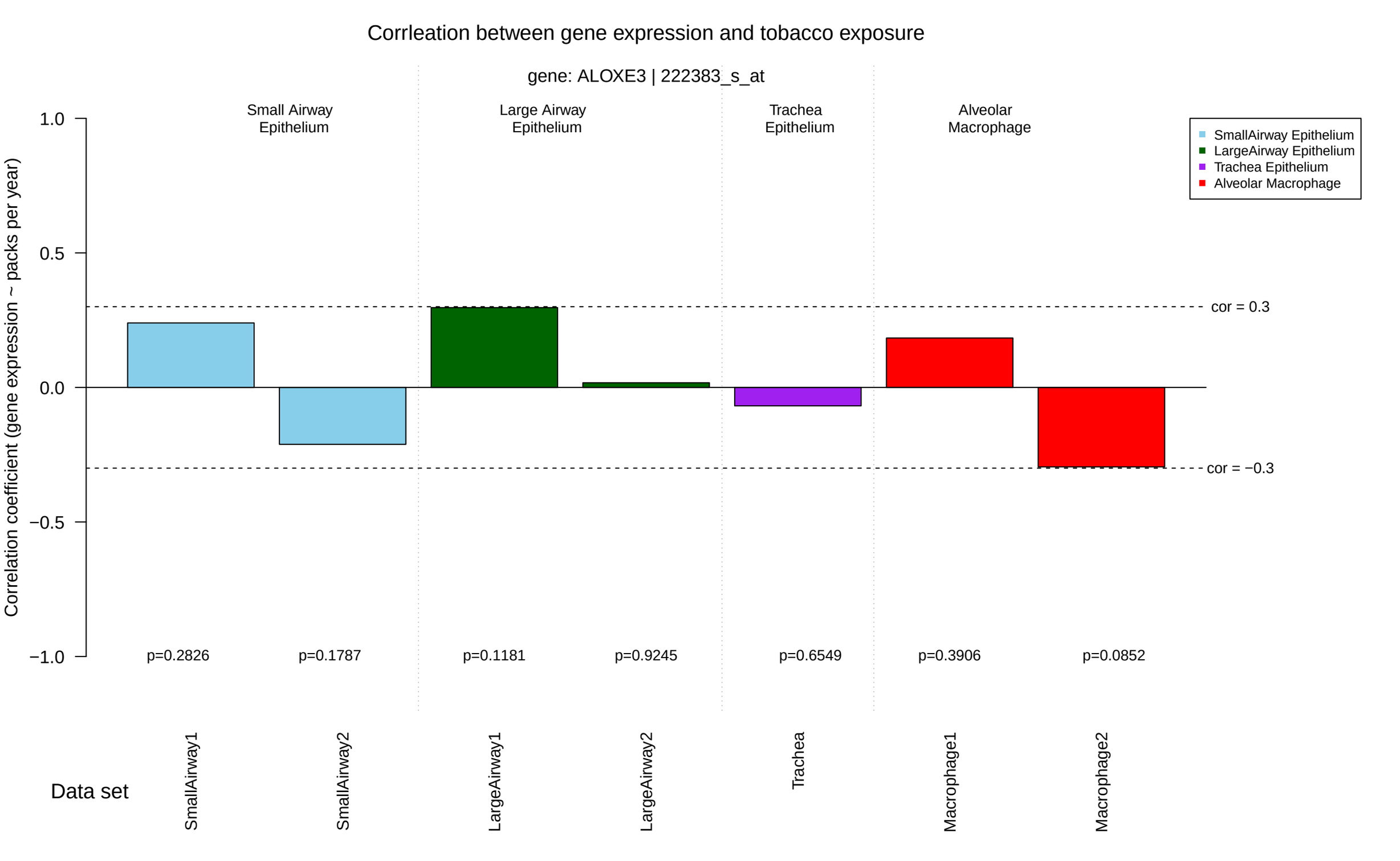

set: LargeAirway1 (p=0.0495)
ALOXE3 | 222383_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5738)
ALOXE3 | 222383_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8063)
ALOXE3 | 222383_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9268)
ALOXE3 | 222383_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2926)
ALOXE3 | 222383_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOXE3 | 222383_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2826
p=0.1787
p=0.1181
p=0.9245
p=0.6549
p=0.3906
p=0.0852
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2